Introduction

Space technology is redefining Africa's future, with AI-driven satellites offering transformative agricultural intelligence to enhance food security and economic growth, advancing self-determined, sustainable development goals. This paper explores how the African Space Agency (AfSA) can expand AI agritech satellites across Sub-Saharan Africa (SSA) to address the region's agrarian challenges, utilizing Kenya as a strategic case study.[1] Framed within the Innovative Systems (IS) Approach, which enables a systemic assessment of institutional and stakeholder dynamics, this analysis identifies essential conditions to foster technological change in a developing economy. As such, Section 1 contextualizes space technologies and AfSA's mission, while Section 2 outlines agricultural challenges across Sub-Saharan Africa. Section 3 explores AI agritech satellite capabilities, followed by an overview of the IS Approach in Section 4. Section 5 applies this framework to Kenya's AI-satellite case study, evaluating its successes and barriers. Lastly, Section 6 synthesizes key case takeaways, and Section 7 discusses research limitations and broader implications for the AfSA's expansion of space solutions to promote sustainable agricultural development in SSA.

1. The Growth of Africa's Space Economy & Advent of AfSA

Once considered a quixotic dream, the timeless aspiration of humanity to explore space has accelerated innovation and discovery on Earth. From Galileo's pioneering telescopic observations to the development of propulsion rockets in the 20th century, the evolution of space technologies is marked by historical milestones. The Cold War-era space race was a catalyst for driving modern space innovation and exponential investment (CTA Staff, 2024). Today, the space industry shows no signs of slowing. According to the Space Economy Report, the global space economy is on a "trillion-dollar trajectory by 2033" (Corrin, 2025). Improvements in navigation, mapping capabilities, and AI imaging are considered key drivers behind the projected market valuation jump of $340 billion over the next few years (Corrin, 2025). Recognized as the next great economic frontier, space not only offers the world access to untapped resources and mechanisms for scientific innovation but also to the terrestrial influence such access would allow. Current trends forecast a new "space race" of frenzied investment to innovate and thus "solve key issues in communication, environment, agriculture, national security" (CTA Staff, 2024). As nations and private enterprises accelerate their pursuit of space-based solutions, space is transitioning

[1] Agritech refers to the "application of tech and digital tools to agriculture to enhance efficiency" (Lutkevich, 2023).

from a mythicized, distant idea to a new arena for economic growth, technological innovation, and geopolitical advantage.

Historically, space innovation has been dominated by Western hegemony. Today, the landscape has shifted. Space technology is increasingly accessible, providing an aspirational avenue for African nations to control their development trajectories outside the Global North's purview. The 21st century "space-power model" is more conducive to "rapid, bottom-up innovation," meaning the industry is more accessible to a wider range of countries, commercial entities, and innovators (Taylor, 2014). The African Union's establishment of the African Space Agency (AfSA) in 2017 marked a turning point in this history, creating governance structures and outlining strategic goals for continental-wide space activities such as "satellite development, space science education, and partnerships with global space stakeholders" (African Space Agency, 2024). Now comprising 55 member states, the AfSA's Statutes underscore the need to promote and coordinate activities that "exploit space technologies and applications for sustainable development and improvement of the welfare of African citizens" (Statute of the African Space Agency, 2018). In particular, the AfSA seeks to "harness the benefits of space science" to address Africa's socio-economic challenges and "develop a sustainable and vibrant indigenous space market that responds to the needs of the continent," while funding space missions to ensure optimal access to spatial data analytics (Statute of the African Space Agency, 2018).

As the global space economy rapidly expands, the AfSA is uniquely equipped to promote the adoption of space technologies for sustainable development in Africa, given its organizational mandate, funding capacity, and partnership network. The organization has enumerated its aspirations to address local challenges including food insecurity and environmental sustainability — and is well-positioned to become a key player in the global space economy as the market landscape evolves. Africa's space sector is on a upward trajectory, currently valued at "$22.64 billion" with substantial growth projected by "2026 due to by strategic initiatives in satellite manufacturing and infrastructure development" (Adetola, 2024). The AfSA can capitalize on this momentum by coordinating power to mobilize space-enabled technologies to promote sustainable development within its collective borders.

One key technology gaining traction and attention in Africa is AI-equipped satellite technologies that can collect precise Earth observation metrics and intelligence to be used for socio-economic initiatives. [2] Africa has sent more than fifty satellites into orbit for agriculture, climate monitoring, disaster management, and telecommunications tracking (Agbetiloye, 2025). Likewise, the AfSA has created new STEM initiatives, like Intelsat, to accrue more technical satellite expertise (SpaceRef,

[2] Earth observation refers to information collected from satellites to monitor Earth's surface and climate (EUSPA, 2024).

2022). Expanding African Earth observation manufacturing segments and advancing Ground Station as a Service models are easing satellite production and reshaping Africa's space ecosystem.[3] By deploying earth observation satellites with AI-driven analytics, African nations are working to strengthen domestic food security, optimize resource management, expand internet access, much of which points the continent towards new mechanisms for sustainable development.

Although Africa's engagement with satellites spans various sectors, I will focus on the application of AI-powered agritech satellites to enhance food security in Sub-Saharan Africa (SSA). While the AfSA operates at a continent-wide level, and the recommendations presented will have broader relevance for sustainable development, my primary focus is SSA. This is due to the region's heavy reliance on agriculture, substantial food security challenges, and the potential for technological interventions to drive impact. Additionally, the case study on Kenya — the leading hub for agritech innovation in Africa — is situated in SSA, making it a more representative example for exploring how the AfSA can integrate AI-powered satellite technologies into agricultural sector in SSA, with potential to iteratively scale across the continent after initial successes.

2. Agricultural Challenges and Opportunities in SSA

As mentioned, the scope of this essay will focus particularly on improving agriculture solutions in SSA using satellite technology. At present, food insecurity is a critical issue affecting the region's political stability and development.[4] SSA is home to more than "950 million people," or "13% of the global population" (OECD, 2016). Despite agriculture being a cornerstone of the region's economy, employing 60% of the workforce, over 218 million people remain undernourished, and the region must import $55 billion dollars of food annually (OECD, 2016). Inefficient resource use, rapid population growth, socio-political conflicts, and civil conflicts have hindered progress towards agricultural optimization. The region also faces persistent challenges in curbing deforestation, managing water scarcity, and mitigating soil degradation (Jamali, 2024). Moreover, the agri-food system remains fragmented as 300 million rural inhabitants spread across remote areas have limited supply networks and inconsistent access agronomic data (Kim et al., 2020). Due to these conditions, SSA is not tracking to meet the UN 2030 goals for Sustainable

[3] GSaaS model provides satellite operators with cloud-based access to ground stations to reduce costs and improve efficiency (Kraetzig, 2024).
[4] The term Sub-Saharan African is defined by the UN Statistical Division and is used to refer to all of Africa, except Northern Africa (UNSD, n.d.).

Development for universal access to safe, reliable food sources (FAO, AUC, ECA, WFP, 2023).

Notwithstanding these persistent challenges, SSA has still witnessed significant growth in the agricultural sector. Between the 1990s and 2013, the total value of agricultural production increased by 130% (OECD, 2016). As Kim et. al note, the "agri-food system can play a key role" in accelerating growth, alleviating poverty and hunger while contributing to job creation in SSA (Kim et al., 2020). With the concurrent expansion of Africa's space market and agricultural sector, these industries can foster synergistic growth, driving innovation in food security strategies while also strengthening regional positioning in the global space economy. More specifically, the AfSA can use AI satellite technology to accrue agricultural intelligence and promote the implementation of sustainable, smart farming in SSA.

To better understand the impact and relevance of AI agritech satellites, I will next outline the broader, foundational benefits of the technology, before delving into Kenya as a case study. Here, Kenya serves as an ideal example due to its proactive adoption of AI-driven agritech satellites and a thriving agricultural innovation ecosystem, making it an advantageous model for other African nations seeking to implement similar initiatives. I will then synthesize insights from Kenya case that the AfSA can leverage to deploy AI satellites across SSA to advance regional goals without adhering to traditional development models characterized by foreign dependencies.

3. Agritech and AI Satellites

This section reviews the current state of satellite technologies as a key component of the agritech portfolio in SSA and explores their role as a disruptive farming technology. While satellites have been used in agriculture for years, they more recently became one of the most widely adopted tools for "remote sensing and precision farming" (Sgargi, n.d.). Satellite technology can provide a comprehensive view of Earth's surface using multispectral imaging, and therefore contribute valuable information of "crop health, vegetation, and soil moisture" (ESA, 2024).[5] In this, satellites can now essentially "track every farm in the world," blending precise contemporary metrics with historical data to illustrate trends in farming outputs (ESA, 2024). These metrics provide agronomists and producers alike with key indicators of agricultural productivity, which offer insights into harvest health and farm management (ESA, 2024).

[5] Satellites help calculate *Normalized Differential Vegetation Indexes* and use *Synthetic Aperture Radar* imaging to develop accurate yield projections (ESA, 2024).

As mentioned, satellites have long been instrumental to Earth observation. The recent integration of AI tools into this practice is refining capabilities by increasing data accuracy, improving real-time processing, and enhancing predictive modeling. Many contemporary satellites come equipped with AI integrations. Here AI permits the autonomous exclusion of non-essential data and ability to "manage mechanical faults, and improve forecasting" analytics for improved operational efficiency (Fearn, 2024a). As the UK Space Agency's Dr. Bahia notes, "AI enables autonomy on satellites," exemplified by the European Space Agency's Φ-sat-1 satellite (Fearn, 2024). The Φ-sat-1 demonstrated AI could successfully filter out poor-quality samples from Earth Observation imagery, ensuring only high-resolution, relevant images were sent to specialists on the ground (Fearn, 2024). Though Kayhan Space's CEO, Dr. Hesar, cations against overestimating the applicability and computational power of AI in space, research from Standford's Food Security scientists indicates satellite imagery and AI are valuable tools to address development data gaps (Sonty, 2023). As the team notes "one satellite image of arable land" can reveal the state of a village's economic health, including crop diversity, yields, and agricultural infrastructure (Burke, 2022).

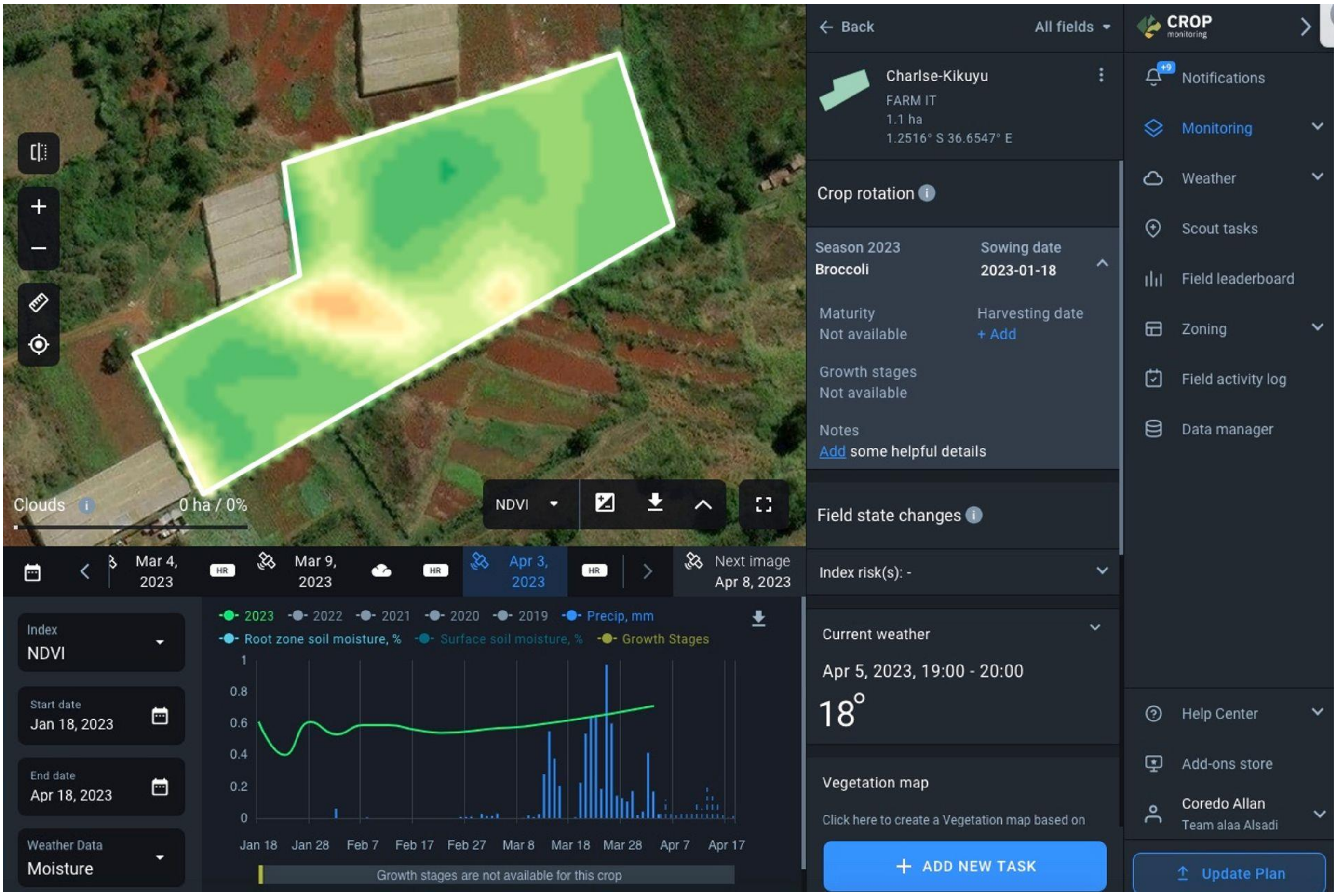

Fig. 1: Interface of AI-satellite platform "FarmIt" in Kenya (Allan, n.d.)

For the discussion of AI satellites in the agritech industry in SSA, I will be referring to the technology as a *disruptive agricultural technology* (DAT). The World Bank group defines DATs as "digital and nondigital innovations that enable smallholder farms to leapfrog constraints, improve yields, incomes, and climate resilience;" AI satellites fall into this categorization ((Kim et al., 2020). For example, AI satellites make farmers more active agents in the food system by providing data insights on pests, disease management, and weather patterns that assist with decision making to optimize yields (Kim et al., 2020). Defining AI satellites as a DAT is also essential to assess their viability because its grounds them within the context of transformative innovations that fundamentally alter traditional farming practices. DAT nomenclature allows for a critical analysis of their scalability, accessibility, and impact on agricultural efficiency. This perspective also allows for a more structured evaluation of their feasibility to diffuse in SSA, as understood through recognized factors such as policy support, infrastructure readiness, and cultural friction. With this context, the next sections will review the Innovation System Approach, before applying the framework to explore Kenya's deployment of agritech satellite as DATs.

4. Innovation Systems Approach

In this section, I will provide a high-level overview of the Innovation Systems (IS) Approach to inform my analysis of Kenya's use of AI agritech satellites. Innovation theories provide structured frameworks for understanding how technologies emerge, spread, and scale within economies. In terms of innovative theories, E.M. Roger's seminal diffusion of innovation (DOI) theory, developed in 1962, is one of the most widely used theoretical frameworks to map technological diffusion. The DOI theory primarily focuses on behavioral dynamics and adoption processes of developed countries, including characteristic determinants for variance in the rate of innovation adoption, rather than the systemic factors that drive innovation (Rogers et al., 2019). On the other hand, the IS Approach, coined by Richard Nelson and Christopher Freeman, has become an increasingly dominant lens in the academic study of innovation and has inculcated considerable influence on policy as such. To provide a more comprehensive analysis of Kenyan agritech innovation and guide the development strategy of the AfSA, the IS Approach offers a more robust to explore the interconnected roles of institutions, policies, and partnerships.

According to IS Approach, innovation involves the "generation and widespread diffusion of knowledge about new products and methods of production" (Lewis, 2021). This occurs via a non-linear process of complex interactions between firms and tangential organizations such as banks, universities, institutes, and governments. According to economist Paul Lewis, innovation here follows the performance of an economy, in the sense that innovation depends critically upon the institutions governing those interactions (Lewis, 2021). In other words, government policy shapes the institutional environment within which innovation can occur. Generally speaking, innovation is "the process whereby new technologies are created" and disseminated throughout the economy to create new "commercially-viable products and novel methods of producing existing goods and services" (Lewis, 2021). The foundational goal is to create new modes of knowledge to increase productivity — meaning it can function as a lever of long-term economic growth. Economists from the World Intellectual Property Organization support this idea, noting "innovation is a driving force of long-term economic growth and sustainable development," especially in states with strong institutional linkages between academic, private, and public sectors (Fu, 2022). The insights of the IS Approach are twofold: innovation is a non-linear process, and this process is shaped by the institutions "governing how firms and other organizations involved in the innovation interact with one another" (Lewis, 2021). To delve into these principles as such:

Non-linear view of innovation

Innovation follows a non-linear trajectory, driven by feedback mechanisms and intertwined relations between science, technology, and production. Users often contribute to the research process to inform innovative learning. Innovation arises from interactions among firms and organizations that provide finance, skills, and information, which then spread throughout the economy. The importance of institutions to govern these interactions leads to the second key principle (Lewis, 2021).

The importance of institutions

The non-linearity of innovation suggests it's shaped by institutional frameworks. Institutions provide infrastructure, policies, and regulations to mediate and facilitate interactions with firms, universities, researchers, and regulators. This structured governance ensures knowledge flows through continuous innovation feedback loops. Thus, systemic orchestration by institutions shapes an environment conducive to innovation, which is especially important in developing countries wherein targeted institutional support can accelerate technological adoption and economic growth (Lewis, 2021).

The IS Approach emphasizes the dynamic and processual nature of economic change, treating change as "occurring endogenously," vis-à-vis to static, neoclassic classifications (Lewis, 2021). Yet critics argue the IS Approach is diffuse, making it difficult to apply both practically and consistently. Acknowledging these epistemic challenges, I still believe the IS Approach still coheres well with the polycentric nature of innovation, supporting the foundational tenets of institutional collaboration and adaptability that allow innovation to diffuse (Lewis, 2021). Institutional innovation spurs sustainable productivity: Kenya's institutions have been integral components of its innovative ecosystem, enabling the country to leverage AI satellites as DATs to enhance productivity and address food security challenges. Lastly, the IS Approach offers a helpful framework to examine the agritech industry as it employs a "holistic view of the adoption decision-making process of farmers" as important stakeholders in the innovation network (Ndah, 2014). Thus, the IS Approach offers an advantageous framework for this paper to understand how Kenya has achieved agritech success with AI satellites — specifically by identifying which "stakeholders, coordinating mechanism, policies" allow for the generation of knowledge and innovation in the agricultural space (Ndah, 2014).

5. Kenya Case Study using IS Approach

Employing an IS Approach, Kenya's implementation of AI agritech satellites highlights strong institutional collaboration across firms, partnerships, and end-users. Over the last two decades, Kenya has emerged as a agritech technology leader in Africa, with one of the most developed digital agriculture ecosystems in SSA (GSMA, 2024). Nearly "30% of all agritech start-ups in SSA" operate in Kenya (Kim et al., 2020). High mobile phone penetration, growing internet connectivity, and a robust mobile money ecosystem have created a strong foundation for the agritech industry. However, the country's supportive institutions, strategic partnerships, dynamic investment, and knowledge accumulation have been key factors to its successful satellite deployment thus far (Kim et al., 2020). Given the agricultural sector is critical to national economy, employing more than a third of the workforce and contributing to "one-fifth of gross domestic product," Kenya has pursued DAT innovation, including satellite technologies, to mitigate the effects of climate change and agricultural disruptions and to help farmers "adopt best practices" to ultimately improve food security (Ombogo, 2023).

5.1.1 Institutional Support

Kenya's agritech sector has benefited from strong institutional support, aligning with the IS Approach, which emphasizes the role of policies, regulations, and knowledge networks in fostering technological advancement. The government has enumerated several regulatory and policy frameworks that prioritize sustainable agritech growth, as seen in Kenya Vision 2030, which integrates digital transformation technologies into agriculture (Kenya Vision 2030, n.d.). Additional initiatives, such as the Bottom-Up Economic Transformation Agenda (2022-2027) and the Nairobi Declaration from the Africa Climate Summit, further promote and guide investment strategies in AI-driven agricultural solutions to enhance climate resilience and productivity (GSMA, 2024). Beyond policy formulation, the Kenyan government actively facilitates collaboration between research institutions, private enterprises, and global development partners, ensuring a vibrant innovation ecosystem. The One Million Farmer Initiative, in conjunction with the World Bank, exemplifies how institutional mechanisms support the adoption of DATs for Kenyan farmers (GSMA, 2024). Furthermore, the proliferation of mobile-enabled agritech services, backed by government initiatives, has accelerated knowledge diffusion and technological integration, reinforcing Kenya's positioning as an innovative agritech leader in SSA. Through structured institutional initiatives, Kenya has created an enabling environment for agribusinesses, researchers, and policymakers to collectively advance AI and satellite-based agritech solutions.

5.1.2 Strategic Partnerships

Kenya's agritech sector demonstrates principles of the IS Approach through strong public-private partnerships, bolstered by institutional support, which drives the adoption for AI agritech satellites. According to Kim, one of the main IS Approach indicators of a strong innovative ecosystem is the openness to disruptive ideas (Kim et al., 2020). World Bank data and stakeholder interviews scored Kenya's innovation ecosystem as highly entrepreneurial and receptive to DATs. Nurturing this societal ethos, Kenya has invested in public-private partnerships to integrate AI-powered satellites into agritech solutions, while acclimatizing farmers to digital solutions. Kenya has strategically solidified technical and investment partnerships, culminating in the launch of its first operational Earth observation satellite, Taifa-1, in 2023 with Kenyan company SayariLabs and external partners Exolaunch and SpaceX (Musambi, 2023.). Collecting greenhouse emission data and hyperspectral images for data and research, Taifa-1 is now likely to be integrated into the SERVIR-Eastern & South Africa, Kenya Ministry of Agriculture, and NASA pilot partnership which aims to empower over 425,000 smallholder farms to better manage climate stresses using satellite-based crop mapping and development methodologies (Ramthun, 2023). Likewise the Agrvision and Directorate of Resource Surveys and Remote Sensing collaboration has further honed the nation's crop monitoring and yield forecasting with a smart unified platform for satellite-based analytics (Farjad, 2023).

5.1.3 Dynamic Investment

The expansion of Kenya's satellite technologies ties directly to a trend of strong government and VC capital infusions into the sector, which has fostered substantial growth and innovation. Kenya is the leading recipient of investments in agritech companies; in 2023, Kenya saw a "45% increase in the share of funding volume captured "in the agritech market on the continent (Briter Bridges, 2024). Kenyan agritech startups exploring agronomic solutions via web apps and digital products, including Twiga Foods, Apollo Agriculture, and Pula, have "pulled a total of

$608 million of funding" in the past decade (Adeyemi, 2024). Significant investment have coincided with higher adoption rates, exemplified by Pula's support for 4 million farmers through satellite-based insurance analytics (*Harvesting Tomorrow*, 2023) These existing platforms have been well-situated to integrate data analytics from satellites to help farmers "make more informed decisions," optimize resources, and adapt to climate variability (Digest, 2023).

5.1.4 Knowledge Accumulation

Kenya's agritech innovation ecosystem, supported by government initiatives, partnerships, and financial investments, exemplifies the IS Approach's non-linear emphasis on knowledge diffusion. Kenyan AI satellite adoption has been reinforced by training programs led by NGOs, research institutions, and government agencies, equipping farmers with real-time satellite data on mobile platforms. The Ajira Digital Program is one such initiative, which trains youth in data science and AI to improve digital literacy (*Ajira*, n.d.). This effort ensures the country's innovation will be self-sustaining into the future and supported by technically skilled workforce. Kenya's agritech sector demonstrates the symbiotic interconnectivity of knowledge between institutions, firms, and end-users that drives iterative, continuous innovations. In another instance of knowledge accumulation, The Ministry of Agriculture, in collaboration with USAID and NASA, routinely gathers informational interviews from farm holders and blends insights with satellite tools on the ground to provide agricultural support (Stephan, 2023). Working with firms and farmers to leverage existing mobile infrastructure and local networks, Kenya's dissemination of agritech has improved digital accessibility.

5.2 Adoption Challenges

Although Kenya's agritech segment is rapidly growing, scaling this innovation and ensuring adoption among isolated farmers remains a challenge (KSA, n.d.). Power shortages, internet connectivity, limited digital literacy, and system interoperability slow the "reach of agritech solutions, especially in rural areas" (*Harvesting Tomorrow*, 2023). In addition, customizing digital platforms to align with local languages and contexts requires adaptive, user-centric iterations that require time and resources to develop. Concerns regarding data privacy, transparency, and equitable access —particularly for women—persist. Thus far, Kenya has a undertaken a holistic approach to address these barriers, and promote inclusive, DAT innovation.

For example, the Kenya Digital Blueprint provides "a conceptual framework" to expand digital access and integrate technologies into the economy (Republic of Kenya, 2019). The Information, Communications, Technology Authority and Investment Authority launched "promotional programs to attract capital and talent" into agritech and satellite development (*Harvesting Tomorrow*, 2023). Its new regulations regarding data protection and cybersecurity address emerging ethical concerns, while tax incentives for tech firms further promote innovation. Infrastructure investments, such as the Konza Technopolis project aim to close gaps in digital literacy (Konza, n.d.). The government has eased the cultural transition to DATs by introducing platforms like iShamba and e-Voucher, which facilitate access to farming inputs for smallholder farms (*Harvesting Tomorrow*, 2023). Lastly, Kenya has implemented several initiatives to address gender inequality in agritech deployments, through policymaking, like The Agriculture Sector Gender Policy, and expansion of extension programs (Waithanji, 2021). Overall, these national initiatives showcase key efforts in Kenya's agritech transformation. As the country expands its use of satellites and other DATs, priorities and barriers will evolve and thus require further evaluation and policymaking to ensure secure and equitable access.

6. Considerations for AfSA

Applying the IS Approach to Kenya's case highlights how the confluence of institutions, firms, and organizations can support an innovative technological system. Kenya's agritech innovation follows a non-linear trajectory — shaped by the interplay between end users, firms, and government agencies — and illustrates the institutional collaboration essential for effective knowledge transfer and accumulation. Additionally, accommodating regulatory frameworks attract DAT innovators, while impact investment capital supports entrepreneurial agritech pilot programs and infrastructure development. Accessible mobile platforms and connectivity accelerate adoption too (Kim et al., 2020). Fig. 2 consolidates the key factors contributing to Kenya's satellite deployment.

IS Approach: Factors Driving Kenya's Success Deployment of AI Agritech Satellites

Factor	Role in Success	Impact
Institutional Strength	• Strong policies & government support for digital agriculture solutions • Strengthens data governance models • Facilitates public-private collaboration	• Enables regulatory approval & scaling of satellite projects (e.g., Taifa-1) • Improves transparency & security in data utilization • Ensures localized policy implementation with relevant governance structures
Strategic Partnerships	• Exemplary collaborations with NASA, SayariLabs, SERVIR, & private firms • Strong relationship between public (universities, research institutes, gov. agencies, local gov.) & private sector (firms, investors)	• Expands access to satellite technology and expertise • Boosts entrepreneurship & multi-sector investments • Improves industrial competitiveness
Dynamic Investment	• Increased venture capital, public funding, & donor investment in agritech solutions • Encourages agritech innovation hubs	• Facilitates infrastructure development & satellite deployment • Promotes research-driven solutions for agritech • Expands market of existing mobile platforms to integrate with satellite data
Knowledge Accumulation	• Digital training programs, AI research integrations, technical courses • Improves digital literacy among farmers • Strengthens agritech research collaborations and promotes secure, open-access platform	• Enhances farmer adoption & sustainable agricultural practices • Inclusion of Indigenous knowledge • Invests in innovation capacity of future generations

Fig. 2: Factors contributing to Kenya's successful satellite deployment

These insights underscore the broader role of institutional quality in supporting innovation diffusion in developing countries. A 2016 Maastricht University study revealed innovation diffusion is shaped by institutional factors, including political, economic, and cultural dimensions. Similarly, Nguyen and Jaramillo's 2004 assessment of 6,000 firms in middle and low-income countries found that lower institutional quality directly correlated to reduced innovation returns (Zanello et al., 2016).[6] Although numerous studies in this field demonstrate technological innovation spurs economic growth, other critics posit innovation can inadvertently exacerbate income disparities, so traditional socioeconomic development tools — such as infrastructure expansion, financial inclusion, and healthcare reforms are preferred (Xiao et al., 2024). In the context of space technologies, Nigerian economists found its space program to be less fruitful than other governmental initiatives in meeting development goals, citing "poor acceptance of space technology" as a major hurdle (Okon, 2018). The nuanced and interdisciplinary nature of innovation studies requires further research and discourse to determine best development practices. However, exogenous patterns that trend toward positive development outcomes can elucidate new mechanisms to guide the expansion of DATs in SSA. My IS analysis of Kenya's agritech sector and satellite deployment illustrates one such example.

To expand AI agritech satellite use in SSA, the AfSA can glean several insights from Kenya's deployment: consistent institutional support, public-private knowledge transfer and investments help catalyze innovative, entrepreneurial ecosystem needed for DATs to thrive. While AfSA's model would operate on a larger scale, applying the IS Approach could facilitate institutional synergy among firms, national governments, and stakeholders. Establishing regional space-technology innovation hubs across SSA could enhance digital education programs, investment initiatives, and technological scalability. Collaboration with local communities, extension officers, and Indigenous knowledge systems to accommodate for cultural and linguistic diversity across the region would ensures cultural inclusivity.

In addition, Kenya has successfully utilized mobile technology and existing digital platforms to provide smallholder farms with accessible satellite-based agricultural insights. AfSA could adopt a similar strategy by investing in rural connectivity and secure data-processing infrastructure. Achieving this effort requires collaborative public-private partnerships with international space agencies to subsidize technological costs, grow adoption, and integrate satellite insights into existing services. Digital Earth Africa, which aims to make Earth observation data

[6] Institutional quality refers to rule of law, regulatory quality, property/patent protections (Zanello et al., 2016).

freely accessible and usable across Africa, presents a viable platform model for broader implementation (Digital Earth Africa, 2020). The table below consolidates potential AfSA entry points to expand AI agritech satellites, inspired by World Bank data visualizations:

Potential Entry Points and Actions for AfSA to Facilitate Adoption of AI Agritech Satellites in SSA		
RECOMMENDATIONS	**SHORT TERM**	**LONG TERM**
POLICY AND REGULATORY REFORMS		
Establish a regional framework for AI-driven satellite applications in agriculture to ensure standardized adoption and regulatory alignment	X	
Develop policies ensuring data sovereignty, accessibility, and ethical governance of satellite-derived agricultural insights		X
Promote national investment incentives for agritech startups leveraging satellite technology		X
INSTITUTIONAL REFORMS		
Strengthening partnerships between AfSA, agricultural ministries, local governments, and private agritech firms to expand satellite use cases	X	
Establish innovation hubs for satellite-driven agritech solutions in high-potential regions	X	
Facilitate farmer training programs through digital literacy initiatives tailored to satellite-based precision agriculture	X	
PUBLIC-PRIVATE INVESTMENTS		
Expand satellite ground station networks to improve real-time agricultural monitoring and climate forecasting		X
Invest in mobile-based applications that translate satellite data into actionable insights for smallholder farmers	X	
Encourage private-sector investment in satellite-enabled agritech startups through targeted funding mechanisms	X	
Foster public-private research partnerships to advance AI-driven precision farming solutions.	X	
KNOWLEDGE SHARING AND CAPACITY BUILDING		
Facilitate knowledge transfer programs between leading agritech nations and emerging markets within Africa	X	
Organize forums and conferences to encourage cross-border collaboration on satellite-driven agritech innovations		X
Support university-led research initiatives focused on the integration of AI and satellite data for sustainable farming	X	

Fig. 3: Recommended entry points/impact timeframe (Kim et al., 2020)

These recommendations for AfSa serve as initial insights, derived from Kenya's specific context. Though they illustrate valuable strategies, a broader, region-wide approach would likely require adaptation to account for regional variations in infrastructure, policy, and technological capacity, as well as addressing systemic challenges. This includes minimizing vestiges of colonial agricultural systems. Export reliance, inequitable land distributions, and foreign economic dependencies have hindered food sovereignty, self-sufficient, and equitable regional development (Roessler et al., 2022). Similarly, African countries have been historically excluded from advanced space research due to Western domination. Progress has been made, but the AfSA could play a crucial role in mitigating these technical and financial barriers such as developing indigenous satellite technologies while supporting local programs, scholarships, and space-tech incubators to encourage local expertise in satellite applications in agritech. Moreover, AfSA can advocate for regional-data sharing agreements, agriculture policies, and African investors to reduce financial and technological reliance on Western entities. By keeping these principles at the forefront of agritech satellite development, AfSA can help lead an inclusive and self-sustaining approach to space-driven agricultural development across the region. Again, these are preliminary considerations; further research is needed to assess real-world feasibility especially as selecting a less agrotechnological developed country as the case study could likely yield different insights.

7. Conclusion

Overall, this paper serves as a normative introduction to the innovation conditions shape Kenya's DAT sector using the IS Approach to conceptualize a framework to guide the AfSA's expansion of AI-satellites on a regional scale. Strong institutional support, knowledge diffusion, strategic public-private collaboration, and investment has helped Kenya develop and deploy AI-satellites to address food insecurity. These factors are not an exhaustive list of considerations for innovation diffusion across SSA, nor are agritech satellites, a panacea for food insecurity. My analysis is inherently qualitative and limited in scope; it does not account for all potential contingencies for agritech innovation, policymaking of this nature relates to a plethora of macroeconomic and social factors excluded from the larger IS Approach. Scaling Kenya's national innovation model into a regional or pan-African context presents its own complexities, due to the spectrum of policy, infrastructure, culture, and technological capacities around the continent. Nonetheless, Kenya's emergence as a regional agritech leader can still help inform the AfSA's regional innovation strategy to expand AI satellite use.

Critics argue investing in space-based agritech solutions may divert resources from more immediate problems such as poverty, inaccessible healthcare, and poverty. Yet, with food security remaining a pressing challenge, space technology can be a "game-changer" (UNOOSA, 2025). Agritech satellites are a valuable DAT that enables farmers to monitor crop health, manage resources, and better safeguard against climate uncertainties. As the continent's leading space agency, AfSA can position space technologies as a new lever for self-determined development in Africa. Unlike other high-tech industries, AI agritech satellites can be regionally led and funded. Traditionally, space exploration and initiatives have been an exclusive endeavor of Western, developed nations. Africa is now rising to be a key player in the industry; with strong institutional architecture and innovation approach as per my analysis, AfSA can use space technology, including satellites, to drive the developmental agenda in SSA. Additional research, policymaking, and strategic frameworks are needed to guide AfSA's pursuit of space technologies as the sector expands; however, these technologies offer Africa a practical tool drive innovation and address development objectives on their own terms.

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